# Habitable Worlds Observatory

HWO is a large infrared/optical/ultraviolet space telescope recommended by the National Academies' Pathways to Discovery in Astronomy and Astrophysics for the 2020s.HWO will be the first telescope designed specifically to search for signs of life on planets orbiting other stars. With a powerful set of instruments, it will also provide a platform for transformational astrophysics.

## A mission that would search for and characterize habitable planets beyond our solar system.

NASA is further prioritizing its long-running search for life in the universe and laying the groundwork for its next flagship astrophysics mission after the Nancy Grace Roman Space Telescope (slated to launch by May 2027.)Currently referred to as the Habitable Worlds Observatory (HWO), HWO builds upon studies conducted for two earlier mission concepts called the Large Ultraviolet Optical Infrared Surveyor (LUVOIR) and Habitable Exoplanets Observatory (HabEx.)

### The Search for Life

Is there other life in the cosmos? How will we know? One telescope will give us the answers to that and more.The Habitable Worlds Observatory: the story of life in the universe.

## Exploring the Trade Space

HWO's engineers perform modeling with various telescope designs to make informed decisions about trade-offs, such as choosing the best balance between telescope size, instrument sensitivity, and stability.

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